Multiclass Classification of Age Groups in the Titanic Dataset

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1. INTRODUCTION

The Titanic dataset has been a benchmark for machine learning applications, with most studies focusing on predicting passenger survival. While survival analysis is widely explored, other perspectives of the data set remain equally valuable. In this project, the focus is shifted to predicting the age group of passengers by categorizing them into predefined classes. This approach may not only provides insights into demographic distribution but also highlights the influence of socio-economic factors on passenger characteristics.

The motivation behind this project is to apply machine learning techniques to a slightly unconventional problem setting using a well-studied dataset, demonstrating that meaningful information can be extracted beyond survival prediction. Furthermore, deploying the model as a Streamlet application allows interactive exploration, enabling users to input hypothetical passenger details and instantly obtain predictions. This bridges the gap between machine learning theory and practical, user-friendly implementation.

2. DATASET

The dataset used in this study is the well-known Titanic dataset, which contains passenger records from the ill-fated RMS Titanic's maiden voyage. It consists of approximately 891 records with features such as passenger class (Pclass), Sex, Age, Number of Siblings/Spouses, Number of Parents/Children, Fare, Port of Embarkation (embarked). Since the Age column in the raw dataset contains missing values, data preprocessing was carried out to handle these gaps. Instead of attempting to predict exact age values, passengers were categorized into five age groups:

- G1: Below 15 years.
- G2: 15 to 30 years.
- G3: 31 to 55 years.
- G4: 56 to 80 years.
- G5: About 80 years.

This transformation allowed the prediction task to be framed as a classification problem rather than a regression problem. The following attributes were used as predictors.

- Pclass – Passenger's class (1,2,3)
- Sex – Male or Female
- SibSp – Number of siblings / spouses aboard
- Parch – Number of parents / children aboard
- Fare – Ticket fare
- Embarked- Port of embarkation (S,C,Q)
- Family Size- SibSp + Parch + 1
- IsAlone - Binary indicator if the passenger traveled alone

These engineered features were added because family structure often correlates with passenger demographics, particularly age.

By carefully curating the data set and applying feature transformations, a more structured and balanced target variable was created, making the classification task feasible and meaningful dynamics.

Figure 2. Distribution of the passengers across different age groups (G1-G5) showing most were between 16-55 years

3. METHODOLOGY

The methodology adapted for this project followed standard steps in machine learning pipeline. First, the data set was cleaned by handling missing values and encoding categorical variables such as Sex and embarked into numerical form. The ticket fare and other contributing factors were Standardize to ensure uniform scale across features.

For model training, the transformed data set was divided into training and testing data subsets. Several Classifiers such h. The models were evaluated on their ability to correctly classify passengers into their respective age groups. The best performing pipeline was then finalized and saved using the Joblib library for deployment.

The next step involved building a user interface with Streamlit. The application allows users to input passenger details such as class, gender, number of family members and ticket fare. On submission, these inputs were converted into a feature vector and passed to the trained pipeline which returns the predicted age group. The design of the application emphasizes simplicity, ensuring that even users without technical expertise can Interact with the system and obtain meaningful predictions.

4. RESULTS

The results of the study showed that the trained model achieved reasonable accuracy in predicting passenger age groups. While exact accuracy metrics depend on the choice of the classifier and the parameter tuning, the overall performance was strong enough to demonstrate that categorical classification of age is feasible using a limited set of sociodemographic features. Notably, the passenger class and fare showed significant correlation with age categories, where younger passengers were often concentrated in the third class, while middle-aged passengers were distributed across all classes.

The interactive Streamlit application successfully integrated the machine learning model into real-time environment. Users could experiment with different combinations of passenger details and immediately observe how the predicted age group varied. This highlighted not only the predictive capability of the model, but also the transparency of the machine learning process.

5. CONCLUSION

This project demonstrates that the Titanic dataset, beyond survival analysis, can also be utilized for predicting demographic categories such as age groups. By categorizing the target variable into 5 meaningful groups and applying feature engineering, the classification task was made more structured and informative. The result confirmed that socio-demographic attributes such as class, fare, and family structure play a key role in predicting age categories.

The integration of machine learning pipeline with Streamlit application further extended the usefulness of the project by providing an interactive platform. Users can now experiment with predictive modeling in hands on manner, making machine learning more accessible. In the future, this project could be expanded by including additional features such as cabin details or text-based attributes, employing advanced classifiers like XGBoost, and visualizing confidence possibilities for each predictive class to provide deeper insights.

The code used to implement this project in this paper is available in the github link as follows: https://github.com/ThanushreeS04/Titanic-MultiClass-Classification/

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